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Service Participation Depth in Experiential Services: Media Touchpoints and Supportive Practices

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Service Participation Depth in Experiential Services: Media Touchpoints and Supportive Practices

Abstract

Purpose

This study examines how media touchpoints, involvement and supportive practices are associated with any live attendance and attendance intensity in spectator sport services.

Design/methodology/approach

Individual-level data from the Sports Life Survey 2024 (Adults) were analyzed using a two-part Bayesian sparse model: a Bernoulli-logit model for any attendance and a zero-truncated negative binomial model for conditional attendance intensity. Prior predictive checks and sensitivity analyses assessed key modeling decisions.

Findings

Any attendance showed positive posterior associations with online J.League viewing, televised Nippon Professional Baseball viewing, transportation support, televised J.League viewing and sport-club involvement. Online Nippon Professional Baseball viewing remained positive across specifications, but its main-model 95% credible interval narrowly included zero. Conditional attendance intensity was positively associated with online J.League viewing and preparation support. Sample-averaged predictions indicated a 22.9-percentage-point difference in any-attendance probability and a 2.71-attendance difference in the conditional expected count for online J.League viewing; preparation support showed a 2.79-attendance difference in the conditional expected count.

Practical implications

Managers should distinguish initiatives aimed at generating physical participation from those aimed at increasing intensity among attendees and should test media-linked attendance communications and coordination-support tools separately.

Originality

The study decomposes service participation into occurrence and conditional intensity and jointly examines mediated touchpoints, involvement and supportive practices within a high-dimensional Bayesian framework.

Keywords: Service participation, customer engagement, experiential services, media touchpoints, attendance

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intensity, Bayesian sparse modeling

Article classification: Research paper

Introduction

Experiential service firms must manage not only whether customers participate, but also how intensively existing participants use the service (Bolton, 1998; Bolton and Lemon, 1999; George and Wakefield, 2018). Customers often encounter experiential offerings through broadcast media, digital platforms and other indirect touchpoints before or between physical service encounters (Lemon and Verhoef, 2016; Kannan and Li, 2017). Mediated and physical consumption may therefore coexist within the same customer relationship rather than represent entirely separate forms of behavior (Kim and Mao, 2021). Nevertheless, empirical service participation is frequently represented through a binary use decision, an aggregate frequency or a longitudinal transition model, leaving the relationship between participation occurrence and conditional intensity insufficiently integrated (Bolton, 1998; Bolton and Lemon, 1999; George and Wakefield, 2018).

In this study, service participation depth is used as an analytical label for two distinct aspects of observed behavior during the previous 12 months: any live attendance and, among attendees, conditional attendance intensity. The concept describes a cross-sectional behavioral structure. It does not imply that the data observe individual customers progressing from non-participation to first attendance or from first attendance to repeated attendance. Participation also refers here to customers' physical use of, or attendance at, an experiential service. It is distinct from customer participation in the production or delivery of a service.

Customer engagement provides a useful theoretical lens because customers' relationships with services extend beyond transactions to include cognitive, emotional, behavioral and social investments (Brodie *et al*., 2011; Vivek *et al*., 2012; Pansari and Kumar, 2017). Customer engagement behaviors can also extend beyond direct consumption through helping, supporting, influencing and mobilizing other actors in a service system (van Doorn *et al*., 2010; Jaakkola and Alexander, 2014). Recent engagement-journey research further emphasizes that engagement occurs across multiple touchpoints, actors and episodes, including contacts outside the firm's direct control (Jaakkola and Alexander, 2024).

The present study does not measure customer engagement as a latent psychological construct. Instead, it examines

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observable media-touchpoint, sport-involvement and support-related indicators that theory identifies as potential manifestations or contextual correlates of engagement. It also includes health, psychological and sociodemographic characteristics as background covariates. These background characteristics are not treated as measures of customer engagement.

Three research streams are particularly relevant. First, customer-journey research explains how customers encounter services through connected mediated and physical touchpoints (Lemon and Verhoef, 2016; Kannan and Li, 2017). Second, service-usage research indicates that whether service use occurs and how intensively a service is used need not reflect identical conditions (Bolton, 1998; Bolton and Lemon, 1999). Third, customer-engagement behavior and value-co-creation research shows that customers and surrounding actors contribute resources through practices extending beyond direct purchase or consumption (van Doorn *et al*., 2010; McColl-Kennedy *et al*., 2012; Jaakkola and Alexander, 2014).

These streams provide important insights but are rarely examined within a common empirical model. Media consumption, physical participation, organizational involvement and support-related practices are commonly studied separately. Consequently, limited evidence exists on whether the same observable indicators are associated with any physical participation and with variation in participation intensity among those who participate.

Spectator sport provides an informative empirical context because mediated consumption, live attendance, club involvement and support-related activities coexist within the same service environment. Customers may follow sports through television or online channels, attend live events with varying frequency, become involved in sport organizations and undertake activities that support other sport participants. The setting therefore permits a broader analysis of how mediated touchpoints and surrounding practices relate to physical service participation.

The study uses individual-level secondary data from a nationwide adult sport survey and applies a two-part Bayesian sparse modeling strategy. The first model examines whether any live attendance occurred during the reference period. The second examines attendance intensity among respondents reporting at least one attendance. A sparse approach is used because each model contains a large number of potentially correlated candidate predictors, while only a comparatively small subset is expected to show appreciable conditional associations.

The study makes three contributions. First, it distinguishes any participation from conditional participation intensity rather than representing attendance through a single behavioral measure. Second, it examines whether mediated touchpoints remain relevant among existing participants instead of treating them solely as pre-participation contacts.

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Third, it directs attention to support-related practices surrounding the focal customer as possible markers of the social embeddedness of experiential-service participation. Bayesian sparse modeling serves as an analytical means of examining these questions rather than as the principal theoretical contribution.

The remainder of the article is organized as follows. The next section develops the theoretical background and research questions. The Method section describes the data, measures, two-part model, sparse Bayesian estimation and model assessment. The Results section reports the posterior evidence, response-scale interpretations and sensitivity analyses. The Discussion considers the theoretical and managerial implications, limitations and directions for future research.

Literature Background

Customer engagement and service participation depth

Customer engagement has become central to services marketing because it extends analysis beyond purchase, satisfaction and loyalty. Foundational research conceptualizes engagement as a multidimensional construct involving cognitive, emotional, behavioral and social investment in interactions with a focal service provider, brand or object (Brodie *et al*., 2011; Vivek *et al*., 2012; Pansari and Kumar, 2017). Customer-engagement behavior research further emphasizes non-transactional actions through which customers affect firms, other customers and broader service systems (van Doorn *et al*., 2010; Jaakkola and Alexander, 2014).

Process-oriented accounts treat engagement as developing across interactions rather than existing as a single static state (Bowden, 2009; Brodie *et al*., 2011). Engagement-journey research similarly argues that customer relationships unfold across multiple touchpoints, actors and episodes (Jaakkola and Alexander, 2024). This perspective is relevant to experiential services because mediated contacts, physical encounters and surrounding social practices may all contribute to the customer's relationship with the service.

The present study focuses specifically on observed physical participation. Participation refers to using or attending an experiential service, rather than participating in its production or delivery. This distinction is important because research on customer participation in services often examines co-production, customer effort or customer roles in service delivery (Bitner *et al*., 1997; Bendapudi and Leone, 2003; Dong and Sivakumar, 2017). Those forms of participation differ conceptually from the attendance behavior examined here.

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Observed participation can itself be represented in different ways. A binary measure distinguishes participation from non-participation, whereas a count measure captures variation in intensity. Combining these aspects into one measure can conceal differences between the occurrence of participation and the extent of participation among users. Service-usage research supports this distinction by showing that the occurrence, frequency and duration of service relationships may have different antecedents and consequences (Bolton, 1998; Bolton and Lemon, 1999).

George and Wakefield (2018), for example, used longitudinal organizational records to model customers' movement from initial purchase to full-season membership in admission-based services. The present study does not observe such within-customer transitions. Instead, it examines a cross-sectional distinction between any live attendance and conditional attendance intensity. The theoretical value lies in showing whether these two components reveal different conditional association profiles, not in claiming to observe an individual progression over time. Sport research has likewise treated repeat attendance as a distinct behavioral outcome and linked it to involvement, loyalty and the sportscape (Hill and Green, 2000).

Media touchpoints and physical participation

Customer experience unfolds across multiple touchpoints before, during and after focal service encounters (Lemon and Verhoef, 2016). These touchpoints may be firm-controlled, partner-controlled, customer-owned or external. Digital marketing research similarly shows that customers interact with offerings through connected digital and physical environments rather than through one linear channel (Kannan and Li, 2017).

Mediated content can support attention, familiarity, knowledge and affective connection. Social-media and digital-content research identifies repeated content contact as one way through which customer engagement may be maintained outside direct transactions (Hollebeek *et al*., 2014; Hollebeek and Macky, 2019). In experiential services, mediated touchpoints may therefore be associated not only with whether customers participate physically, but also with how intensively participants remain involved.

Spectator sport makes this relationship especially visible. Customers can follow leagues, teams, athletes and competitive narratives through television and online channels without attending in person. Kim and Mao (2021) show that live attendance and mediated sport consumption reflect distinct but connected consumption forms. Media consumption may increase familiarity, service salience and knowledge of schedules and participants, which may be associated with physical attendance.

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The relationship need not be uniformly complementary. Event-level attendance-demand research has reported instances in which live broadcasting substitutes for stadium attendance. Allan and Roy (2008), for example, found a reduction among pay-at-the-gate home supporters when Scottish Premier League matches were broadcast live. A wider scoping review documents the breadth and heterogeneity of stadium-attendance demand research across sports and settings (Schreyer and Ansari, 2022). These findings do not necessarily conflict with a positive individual-level association between habitual media use and attendance. Broadcasting a particular event may reduce attendance at that event, whereas customers who routinely consume sport media may also be more involved, informed or motivated overall.

The relevant question is therefore not whether media and physical participation are always complements or substitutes. It is whether particular media behaviors show conditional associations with any physical attendance, with attendance intensity among attendees, or with both components.

Supportive practices and socially embedded participation

Customer-engagement behavior and value-co-creation research provides a basis for examining actions surrounding the focal service encounter. Customers can affect a service system through helping, supporting, influencing and mobilizing others, even when those actions are not direct purchases or focal consumption behaviors (van Doorn *et al*., 2010; Jaakkola and Alexander, 2014).

Service-dominant logic similarly views customers and other actors as resource integrators who combine time, effort, knowledge, relationships and practical activities in value creation (Vargo and Lusch, 2008, 2016). Practice-based research further emphasizes that service-related value is embedded in customers' everyday activities and social contexts (McColl-Kennedy *et al*., 2012).

Consumption is also frequently organized through families, households and other social units rather than isolated individuals. Epp and Price (2008) show how consumption practices contribute to family identity, while Letheren *et al*. (2019) conceptualize household engagement as involving coordinated roles and resources. These perspectives suggest that participation in experiential services may be associated with activities occurring around the focal customer.

The secondary survey used in this study includes indicators such as transporting sport participants and preparing drinks or meals for them. These items do not constitute validated measures of household engagement. Nor do they

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directly measure assistance with live attendance. They are treated as specific observable support-related practices that may mark a broader sport-active social context.

Such practices could be associated with participation for several reasons. They may keep sport salient in everyday life, indicate involvement in sport-related social networks or reflect routines in which sport activities are already integrated. Alternatively, the association may reflect selection: individuals who are already highly involved in sport may be more likely both to attend events and to support other participants. The cross-sectional analysis cannot distinguish these mechanisms, but it can assess whether such indicators remain associated with participation after a broad range of other observed characteristics is considered.

Integrative gap and research questions

Existing research establishes that customer engagement spans multiple touchpoints and actors, that physical service usage can vary in occurrence and intensity, and that support-related practices may be embedded in customers' social contexts. These insights, however, are commonly investigated in separate research streams.

Customer-engagement and customer-journey research often theorizes multiple touchpoints but does not distinguish the conditional correlates of any physical participation from those of participation intensity. Sport-consumption research examines media use, identification, servicescape and attendance, but frequently focuses on already-engaged fans or a relatively narrow predictor set. Value-co-creation research provides a basis for interpreting supportive practices, but such practices are rarely examined alongside mediated consumption and live attendance within a common empirical model.

This fragmentation leaves two questions unresolved. First, it is unclear which media-touchpoint, involvement and support-related indicators are associated with the occurrence of any live attendance. Second, it is unclear which indicators are associated with variation in attendance intensity among those who attend.

The secondary dataset contains a broad range of candidate indicators, while the existing literature does not provide a sufficiently precise basis for assigning each individual item to one component in advance. The study therefore adopts exploratory research questions rather than directional hypotheses:

RQ1. Which media-touchpoint, involvement and supportive-practice indicators are associated with any live attendance during the 12-month reference period?

RQ2. Which indicators are associated with attendance intensity among attendees, and how does the posterior evidence

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profile compare with that for any attendance?

Method

Data source and sample

This study uses individual-level secondary data from the Sasakawa Sports Foundation's Sports Life Survey 2024 (Adults), the adult sample of the Sports Life Data 2024 series (Sasakawa Sports Foundation, 2024). The Sasakawa Sports Foundation is a Japanese nonprofit research organization that conducts recurring nationwide surveys of sport participation and sport-related life.

The survey was planned by the Sasakawa Sports Foundation, and fieldwork was conducted by Nippon Research Center. It covered men and women aged 18 years or older living in municipalities across Japan. Quota sampling was used across 300 sampling points. The survey was administered as a drop-off and pick-up paper questionnaire from June 6 to July 7, 2024. The original sample contained 3,000 respondents.

The questionnaire included items on personal sport participation, direct sport spectating, television and internet sport viewing, volunteering, sport-club involvement, support activities, health, lifestyle and sociodemographic characteristics. J.League refers to Japan's professional football leagues, and Nippon Professional Baseball (NPB) refers to Japan's top professional baseball league.

After cases with incomplete values were removed, 2,888 respondents remained. Predictors with no variation were removed separately during preprocessing. One record was then excluded because its binary attendance response indicated attendance while its reported attendance count was zero. The final any-attendance sample therefore contained 2,887 respondents. Of these, 762 reported at least one live attendance and formed the analytic sample for the conditional-intensity model.

The data were obtained through the Sasakawa Sports Foundation's formal data-application procedure. The analysis used no direct personal identifiers.

Measures and preprocessing

The first outcome was any live attendance, defined as whether the respondent reported at least one live sport attendance during the previous 12 months. Although the dataset contained a separate binary attendance item, the

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outcome was derived directly from the reported attendance count:

$$A_i = 1(Y_i > 0),$$

where Y_i denotes the number of live sport attendances reported by respondent i. Thus, $A_i = 1$ indicates at least one attendance during the reference period, and $A_i = 0$ indicates no attendance. This measure does not identify a respondent's first-ever live attendance.

The second outcome was conditional attendance intensity, defined as the number of live sport attendances among respondents with $Y_i > 0$. The outcome includes respondents who attended exactly once and therefore represents variation in positive attendance frequency rather than a direct measure of transition to repeat attendance.

Candidate predictors were organized into focal behavioral indicators and background covariates. Focal indicators represented mediated sport touchpoints, sport-related involvement and supportive practices. Television and online viewing represented mediated touchpoints. Sport-club, volunteering, transportation and preparation activities represented involvement or support-related practices. Health, psychological state and sociodemographic variables were included as background covariates that could account for alternative sources of variation in attendance. These background characteristics were not treated as measures of customer engagement.

Before estimation, variables with no variation were removed. Two "none of the above" indicators were also excluded: x366, indicating no relevant application or game use, and x580, indicating no relevant sport-club or support experience. The absence of each specific behavior already represented non-use or non-experience, making the summary indicators redundant for interpretation.

Predictor matrices were screened for exact linear dependencies. When an exact dependency was identified, the later variable in the source-variable order was removed. One redundant indicator was removed from each model component; neither was among the posterior associations reported in the article. The resulting any-attendance model contained 262 predictors. After the sample was restricted to attendees and variables without variation in that subsample were removed, the conditional-intensity model contained 215 predictors.

In the primary specification, continuous and numerically coded ordered predictors were standardized using means and standard deviations calculated in the full analytic sample. The same scaling constants were used in both model components. Binary indicators remained coded 0/1.

Standardization was a one-to-one change in measurement units and did not remove observations, categories or rank

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information. Scaling was relevant because the implications of a common shrinkage prior depend on predictor units. A sensitivity analysis therefore used an alternative preprocessing specification in which continuous variables were divided by two standard deviations and ordered categorical variables were represented by indicator variables rather than assumed to have equally spaced categories (Gelman, 2008). After exact-alias removal, the alternative-preprocessing models contained 294 and 247 predictors, respectively.

Two-part modeling strategy

The two-part strategy separates two behavioral questions. The first component models whether any live attendance occurred during the reference period. The second models attendance frequency among respondents who reported at least one attendance. This decomposition is cross-sectional and does not estimate within-person transitions from non-attendance to first attendance or from first to repeated attendance.

The first component was specified as a Bernoulli-logit model:

$$A_i \sim Bernoulli(\pi_i),$$

$$logit(\pi_i) = \alpha_1 + X_i\beta_1.$$

In plain terms, this component asks whether a respondent reported any live attendance during the reference period.

The second component modeled positive attendance counts using a zero-truncated negative binomial (ZTNB) model:

$$Y_i \mid Y_i > 0 \sim ZTNB(\mu_i, \varphi),$$

$$log(\mu_i) = \alpha_2 + X_i\beta_2.$$

Because the conditional-intensity sample contains only positive counts, a zero count is impossible by construction. The likelihood therefore conditions the negative binomial distribution on $Y_i > 0$ (Mullahy, 1986; Grogger and Carson, 1991; Cameron and Trivedi, 2013):

$$p(Y_i = y_i \mid Y_i > 0) = pNB(y_i \mid \mu_i, \varphi) / [1 - pNB(0 \mid \mu_i, \varphi)], \qquad y_i = 1, 2, \ldots$$

The negative binomial distribution accommodates variation in positive attendance counts beyond that permitted by a Poisson distribution. The zero-truncated specification differs from a zero-inflated model: a zero-inflated model assumes that observed zeros may arise from multiple latent processes, whereas the present design directly separates no attendance from positive attendance frequency.

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Exponentiated count-model coefficients refer to ratios on the underlying negative-binomial μ scale. Because the outcome is zero-truncated, they are not interpreted as simple ratios of observed positive counts. Response-scale expected positive counts were calculated as:

$$E(Y_i \mid Y_i \geq 1) = \mu_i / [1 - P(Y_i = 0)].$$

Sparse Bayesian estimation

A Bayesian sparse approach was selected because the research questions required the simultaneous examination of large sets of potentially correlated predictors—262 in the any-attendance model and 215 in the conditional-intensity model—under the expectation that only a comparatively small subset would show appreciable conditional associations.

An unregularized model containing this many predictors could yield unstable estimates and overfit the observed sample. Frequentist penalized estimators provide viable alternative forms of regularization. The regularized horseshoe was selected because it represents sparsity explicitly while providing joint posterior distributions for every coefficient and for response-scale predictions. The choice was therefore tied to the high-dimensional and exploratory character of the analysis rather than to a claim that Bayesian methods are generally superior to frequentist methods.

In practical terms, a regularized horseshoe prior applies strong shrinkage to coefficients with limited empirical support while permitting coefficients supported by the data to remain farther from zero (Carvalho *et al*., 2010; Piironen and Vehtari, 2017; van Erp *et al*., 2019). It does not make a deterministic in-or-out selection decision. Every coefficient retains a posterior distribution, which permits uncertainty to be summarized through credible intervals, posterior probabilities and predictive contrasts.

For both model components, the expected number of non-negligible coefficients was set to $p_0 = 10$. The expected nonzero-to-zero coefficient ratio used to calibrate global shrinkage was:

$$par_ratio = p_0 / (p - p_0),$$

where p denotes the number of coefficients in the relevant component. The value $p_0 = 10$ represented a deliberately sparse working assumption that only a small minority of the candidate indicators would show appreciable conditional associations. Because this number was uncertain, sensitivity analyses used $p_0 = 5$ and $p_0 = 20$.

The any-attendance coefficients used a regularized horseshoe prior with local degrees of freedom of 3, a slab scale

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of 2 and slab degrees of freedom of 4. The centered any-attendance intercept was assigned a *Normal(0, 2.5²)* prior. The joint implications of these settings were evaluated through prior predictive simulation, which produced a deliberately broad distribution over feasible attendance rates (see web appendix, Figure S1).

Prior predictive checking was used to calibrate the conditional-intensity priors (Gabry *et al*., 2019). Initial checks indicated that less regularizing specifications generated an excessively heavy upper tail. Candidate specifications were therefore evaluated sequentially using criteria based only on the marginal prior-predictive count distribution, rather than on coefficient estimates or the identity of the reported predictors.

The final conditional-intensity specification used local degrees of freedom of 3, a slab scale of 0.5 and slab degrees of freedom of 100. The centered intercept was assigned:

$$\alpha_2 \sim Normal(\log(4.147), 0.5^2),$$

where 4.147 was the observed mean positive attendance count. This is therefore a data-scaled weakly informative prior rather than a prior derived entirely from external information. The negative-binomial shape parameter was assigned a Gamma prior with shape 4 and rate 4:

$$\varphi \sim Gamma(4, 4).$$

The final specification retained the observed count summaries within broad prior-predictive ranges while placing substantially less mass on implausibly extreme annual counts. The prior-predictive distributions under the final specifications are shown in the web appendix (Figure S1).

Models were estimated in R version 4.5.3 using brms version 2.22.0 (Bürkner, 2017, 2018) with the cmdstanr 0.9.0 backend and CmdStan version 2.38.0. Stan implements Bayesian inference through Hamiltonian Monte Carlo (HMC), using the No-U-Turn sampler (NUTS) to adapt path lengths (Carpenter *et al*., 2017; Hoffman and Gelman, 2014). Each posterior model used four chains, 4,000 iterations per chain and 2,000 warm-up iterations, yielding 8,000 post-warm-up draws. Sampling used adapt_delta = 0.99, max_treedepth = 15 and random seed 12345.

Model assessment, posterior interpretation and sensitivity analyses

Model assessment followed a Bayesian workflow comprising prior predictive checks, convergence diagnostics, posterior predictive checks and approximate leave-one-out cross-validation.

Convergence was evaluated using the rank-normalized R-hat statistic, bulk effective sample size and tail effective

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sample size (Vehtari *et al*., 2021), together with divergent transitions and maximum tree-depth diagnostics.

Posterior predictive checks examined whether the fitted models reproduced the observed attendance rate and key features of the positive-count distribution, including the mean, variance, median and upper quantiles. Predictive performance and influential observations were evaluated through Pareto-smoothed importance-sampling leave-one-out cross-validation (PSIS-LOO; Vehtari *et al*., 2017, 2024).

For concise presentation, coefficients whose 95% credible intervals excluded zero under the final main specification are treated as the principal reported posterior associations. Online NPB viewing is also reported separately because its posterior median remained positive and similar across all specifications, although its final main-model 95% credible interval narrowly included zero.

For each reported coefficient, the analysis provides the posterior median, 95% credible interval (CrI), posterior probability of a positive coefficient and exponentiated coefficient. For selected binary indicators, average response-scale predictive contrasts were also calculated. The indicator was set to 0 and 1 for every respondent in the relevant analytic sample; predictions were averaged across respondents within each posterior draw and then summarized across draws. These quantities are model-based, sample-averaged adjusted associations rather than causal effects.

Sensitivity analyses examined more restrictive and less restrictive sparsity assumptions ($p_0 = 5$ and $p_0 = 20$), alternative predictor preprocessing and exclusion of the single respondent reporting the maximum of 50 attendances. The complete-sample count prior was held fixed in the maximum-count exclusion model so that only the observation changed.

Results

Model assessment and predictive checks

Both model components showed satisfactory sampling diagnostics. Across the main models, the maximum R-hat among all saved parameters was 1.003, and no divergent transitions or maximum tree-depth hits occurred. Across all saved parameters, the minimum bulk and tail effective sample sizes exceeded 1,800.

The maximum Pareto-k values were 0.633 for the any-attendance model and 0.637 for the conditional-intensity model. Both values were below 0.7; exact leave-one-out refitting was therefore not required.

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Posterior predictive checks reproduced the observed any-attendance rate of 0.264. For conditional attendance intensity, the observed mean was 4.147, while the posterior median of the replicated means was 4.202. The observed median was 2, and the posterior median of the replicated medians was also 2. The observed 90th, 95th and 99th percentiles were 10, 14 and 32, compared with posterior medians of 9, 13 and 28.78 for the corresponding replicated percentiles. The model therefore reproduced the center and principal upper quantiles reasonably well, although the posterior median of the replicated 99th percentile was modestly below the observed value.

The estimated negative-binomial shape parameter was 0.447 (95% CrI: 0.294 to 0.634), indicating substantial extra-Poisson variation in positive attendance counts.

Posterior evidence for any live attendance

Five coefficients met the principal reporting criterion in the any-attendance model. The clearest positive posterior associations were observed for online J.League viewing, televised NPB viewing and transportation support. Televised J.League viewing and sport-club involvement also had positive coefficients, although their lower credible-interval bounds were close to zero.

Online NPB viewing showed a stable positive posterior median across all specifications. In the final main model, however, its 95% credible interval narrowly included zero ($\beta = 0.425$, 95% CrI: −0.001 to 0.886), although the posterior probability of a positive coefficient was 0.974. This association is therefore treated as suggestive rather than as a principal finding.

Online J.League viewing was reported by 83 respondents (2.9%) in the full analytic sample. The width of its credible interval should be interpreted in light of this relatively low prevalence.

[Insert Table I about here]

Posterior evidence for conditional attendance intensity

Two coefficients met the reporting criterion in the conditional-intensity model: online J.League viewing and the preparation-support indicator.

Online J.League viewing had a posterior median coefficient of 0.703 (95% CrI: 0.240 to 1.122; $P(\beta > 0) = 0.997$). The preparation-support indicator had a posterior median coefficient of 0.785 (95% CrI: 0.380 to 1.171; $P(\beta > 0) > 0.999$).

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Online J.League viewing was reported by 59 attendees (7.7%), while the preparation-support behavior was reported by 127 attendees (16.7%). The preparation-support item refers to preparing drinks or meals for sport participants. It does not directly measure support for live attendance and is not interpreted as a complete measure of household engagement.

The single respondent reporting 50 attendances was excluded in an influential-case sensitivity analysis. The coefficients remained positive for online J.League viewing ($\beta = 0.719$, 95% CrI: 0.264 to 1.147) and preparation support ($\beta = 0.705$, 95% CrI: 0.304 to 1.085). The findings were therefore not dependent on the highest-frequency attendee.

[Insert Table II about here]

Response-scale interpretation and sensitivity analyses

Response-scale predictions provide a more direct interpretation of the main posterior associations. After adjustment for the remaining predictors, the sample-averaged posterior median probability of any attendance was 25.7% when online J.League viewing was set to 0 and 48.6% when it was set to 1. The corresponding difference was 22.9 percentage points (95% CrI: 10.1 to 36.0).

Among attendees, the expected positive attendance count was 3.95 when online J.League viewing was set to 0 and 6.66 when it was set to 1. The difference was 2.71 attendances (95% CrI: 0.75 to 5.37), and the conditional count ratio was 1.69 (95% CrI: 1.19 to 2.37).

For preparation support, the expected positive attendance count was 3.59 when the indicator was set to 0 and 6.38 when it was set to 1. The difference was 2.79 attendances (95% CrI: 1.21 to 4.75), and the conditional count ratio was 1.77 (95% CrI: 1.31 to 2.38).

These quantities are model-based, sample-averaged adjusted associations from observational data and should not be interpreted as causal effects.

[Insert Table III about here]

The seven principal coefficients retained their direction and substantive magnitude under $p_0 = 5$, $p_0 = 20$ and the alternative preprocessing specification. Online NPB viewing remained positive and similar in magnitude in every specification, although its final main-model interval narrowly included zero (see web appendix, Table S1).

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The alternative preprocessing specification produced two additional positive coefficients—televised rugby viewing and one level of the sport-participation variable—that did not emerge under the primary preprocessing. These specification-specific coefficients are not interpreted as principal findings.

Overall, the two model components yielded different posterior evidence profiles. The any-attendance model showed evidence across several media touchpoints, transportation support and club involvement, whereas the conditional-intensity model showed evidence for online J.League viewing and preparation support. Online J.League viewing appeared in both components.

These comparisons describe the posterior evidence obtained from each model. They do not constitute formal tests that corresponding coefficients differ across components because the models use different outcomes, link functions, samples and predictor sets. The narrower evidence profile in the conditional-intensity model should also be interpreted in light of its smaller analytic sample and the shrinkage applied to a high-dimensional predictor set.

Discussion

Overview of findings

This study examined any live attendance and conditional attendance intensity as two components of service participation depth. The two components produced partly overlapping but different posterior evidence profiles.

Any live attendance was associated with several media-touchpoint indicators, transportation support and sport-club involvement. Online J.League viewing showed the largest posterior association and also remained positively associated with attendance intensity among attendees. Conditional attendance intensity was additionally associated with the preparation-support indicator.

These results represent conditional associations in cross-sectional data. They do not demonstrate that media viewing or support activities cause attendance, nor do they show individual customers progressing from non-attendance to first or repeated attendance.

Theoretical implications

Decomposing service participation depth

The first contribution is to distinguish the occurrence of physical participation from variation in intensity among

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participants. Service research has long recognized that usage and relationship duration have dynamic and multifaceted antecedents and consequences (Bolton, 1998; Bolton and Lemon, 1999). Nevertheless, empirical participation is often represented through a single binary outcome, an aggregate frequency or an assumed progression from trial to continued use.

The present decomposition indicates that any attendance and conditional attendance intensity need not produce identical posterior evidence profiles. The any-attendance model yielded a broader set of media, transportation and organizational indicators. The conditional-intensity model yielded a narrower profile centered on online J.League viewing and preparation support.

The broader any-attendance profile also differs from earlier repeat-attendance research that emphasized involvement, loyalty and the sportscape (Hill and Green, 2000). In the present high-dimensional model, sport-club involvement appeared in the any-attendance component rather than the conditional-intensity component. This contrast is descriptive, because the two components were not formally compared and the studies use different measures and settings.

This contribution is narrower than claiming to observe a customer journey. George and Wakefield (2018) modeled longitudinal movement from initial purchases toward full-season membership using organizational records. The present cross-sectional data do not identify such transitions. Instead, the results indicate that a single attendance measure can conceal different conditional association structures for attendance occurrence and positive attendance frequency.

The narrower conditional-intensity profile may reflect a more selective pattern of association, but it may also partly reflect the smaller analytic sample, greater posterior uncertainty and the different information-to-parameter ratio in that component. The theoretical contribution therefore lies in the value of separating the two components, not in claiming that each difference between reported coefficients represents a true component-specific causal effect.

Media touchpoints across participation components

The second contribution concerns the role of mediated touchpoints. Customer-journey research emphasizes that service relationships span contacts before, during and after physical encounters (Lemon and Verhoef, 2016). Digital-content research similarly indicates that mediated contacts can sustain attention, familiarity and activation outside direct transactions (Hollebeek *et al*., 2014; Hollebeek and Macky, 2019).

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Online J.League viewing showed posterior evidence in both model components. The response-scale results show a 22.9-percentage-point difference in any-attendance probability and a 2.71-attendance difference in the conditional expected count after adjustment for the remaining predictors. This pattern is consistent with the view that mediated and physical sport consumption are distinct but connected forms of behavior (Kim and Mao, 2021).

Several mechanisms could generate this association. Repeated online viewing may maintain sport salience, increase familiarity with teams and schedules, reduce uncertainty about the service or reinforce sport-related routines. These mechanisms were not directly measured and should be treated as testable interpretations rather than demonstrated mediators.

Reverse causality is equally plausible. Individuals who attend frequently may subsequently consume more online content. General sport involvement or identification may also contribute to both behaviors despite the broad set of observed covariates.

The result must also be interpreted in relation to the wider attendance-demand literature. Allan and Roy (2008), for example, found a reduction among pay-at-the-gate home supporters when Scottish Premier League matches were broadcast live. A positive individual-level association is not necessarily inconsistent with this result. Broadcasting a specific event may substitute for attending that event, whereas habitual media users may be more engaged, informed or involved overall.

Finally, online J.League viewing is league-specific, whereas the outcome measures attendance at any live sport event. The finding should therefore be described as an association between a specific media behavior and broader live-sport participation, not as evidence that online J.League viewing converts customers specifically into J.League attendees.

Preparation support and socially embedded participation

The third contribution is to direct attention toward practices surrounding the focal customer. Preparing drinks or meals for sport participants is neither live attendance itself nor a validated measure of household engagement. It is best interpreted as one observable preparation-support practice that may mark a broader sport-active social context.

This interpretation extends empirical research showing that consumption and engagement can be organized through coordinated family or household roles rather than through isolated individual action (Epp and Price, 2008; Letheren *et al*., 2019). The present result differs in that the support item is not a measure of household engagement and does not identify the recipient as a household member. It nevertheless suggests that an observable support-related practice

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can be associated with the intensity of physical service participation.

The response-scale results show a 2.79-attendance difference in the conditional expected count after adjustment for the remaining predictors. The result remained positive after the highest-frequency attendee was removed and under alternative sparsity and preprocessing specifications.

One possible interpretation is that sport-related roles and routines embed sport more deeply in everyday life. Preparing food or drinks for participants may indicate that sport is regularly discussed, scheduled or coordinated within the respondent's social environment. Such a context may maintain sport salience and create more opportunities for live attendance.

Selection provides an important alternative explanation. Individuals who are already highly involved in sport may be more likely both to attend events frequently and to support other participants. The item may also reflect gendered household roles, caregiving responsibilities or support for persons outside the household. The data cannot distinguish these possibilities.

The theoretical implication is therefore not that meal preparation directly produces attendance. Rather, the result suggests that research on experiential-service participation should consider actors and practices surrounding the focal customer, particularly when participation is embedded in recurring social routines and requires coordination with others.

Managerial implications

The findings suggest that managers should distinguish between increasing the occurrence of any physical participation and increasing attendance intensity among existing attendees.

For any attendance, mediated touchpoints can be connected more directly to low-friction participation pathways. Streaming pages, highlights, previews and team-related digital content can provide links to event schedules, venue information, access instructions, seating guidance and introductory attendance offers. Relevant performance indicators include subsequent attendance among media-engaged customers, use of information designed for prospective attendees and abandonment before ticket purchase.

For attendance intensity, managers can use post-event and between-event communication to maintain salience and reduce the effort required to plan subsequent attendance. Relevant practices include personalized event reminders, flexible multi-event bundles, calendar integration, transport and parking guidance and group-planning functions.

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Appropriate performance indicators include annual attendance frequency and lapse rates among previous attendees.

The preparation-support result also suggests the value of designing around customers' social context. Service organizations cannot directly manage household or social routines, and the survey item should not be used as a targeting criterion. Managers can, however, reduce coordination burdens by offering family or group booking functions, food-preorder options, clear information for accompanying persons and communication tools that can be shared among participants and supporters.

These recommendations should be evaluated through field experiments or phased implementation because the present results are associational. Similar distinctions between participation occurrence and intensity may be relevant to theaters, museums, festivals, theme parks and other recurring experiential services that combine mediated and physical touchpoints.

Limitations and future research

Several limitations define the scope of the findings.

First, the data are cross-sectional. The analysis identifies conditional associations but cannot establish causal ordering. Media consumption may precede physical attendance, but attendance may also increase subsequent media use. Preparation support may facilitate participation, reflect an already sport-active social context or arise as a consequence of sport involvement. Longitudinal panels, linked behavioral records and field experiments are required to establish temporal ordering and causal mechanisms.

Second, the two outcomes capture behavior within a 12-month reference period. Any live attendance does not identify a respondent's first-ever attendance. Conditional attendance intensity includes respondents who attended exactly once and therefore does not directly estimate the transition from one-time to repeated attendance. Future research should observe actual transitions from non-attendance to first attendance and from first attendance to subsequent participation.

Third, attendance frequency is self-reported and may be affected by recall error. Linking media-exposure records, ticket transactions and venue-entry records would provide more precise behavioral measures.

Fourth, the secondary survey was not designed to measure the full cognitive, affective or social structure of customer engagement. The included variables are observable indicators and background covariates rather than psychometric measures of engagement. The preparation-support item is a single binary measure referring to support for sport

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participants, not direct support for live attendance. Future research should develop richer measures of transportation, scheduling, emotional support, financial support, caregiving coordination and shared social routines.

Fifth, the media indicators and attendance outcome are not perfectly matched by service object. J.League and NPB media use are related to attendance at any live sport event. Future studies should connect league-, team- or event-specific media exposure to corresponding attendance behavior.

Sixth, online J.League viewing was relatively uncommon in the sample. Although posterior uncertainty reflects its low prevalence, replication with larger numbers of online viewers would strengthen confidence in the magnitude of the association.

Seventh, the Japanese spectator-sport context may limit generalizability. Household structure, gender roles, media availability, event frequency, transport requirements and sport institutions vary across countries and service categories. Replication is needed in other countries and in other experiential services.

Eighth, quota sampling limits population-level generalization. The results characterize conditional associations in the analytic sample and should not be interpreted as design-based national population estimates.

Finally, comparisons between the two components are descriptive rather than formal coefficient-difference tests. The models use different outcomes, links, samples and predictor sets. The smaller conditional-intensity sample may also yield greater posterior uncertainty and a different pattern of regularization. Sensitivity analyses reduce concern that the principal findings are artifacts of one sparsity assumption or coding choice, but they do not remove all model dependence.

Future research can address these limitations through longitudinal customer-level data, service-object-matched media and attendance records, dyadic or household data, multi-item measures of supportive practices and field interventions designed to test media-linked attendance communication and coordination-reduction mechanisms.

Conclusion

This study examined service participation depth by distinguishing any live attendance from conditional attendance intensity among attendees. A two-part Bayesian sparse model was used to estimate conditional associations between a broad set of media-touchpoint, sport-involvement, supportive-practice and background indicators and the two participation components.

Any live attendance showed posterior associations with several media touchpoints, transportation support and sport-

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club involvement. Online J.League viewing showed evidence in both components. Conditional attendance intensity was additionally associated with a preparation-support indicator. Sample-averaged predictions associated online J.League viewing with a higher probability of any attendance and greater attendance intensity, while preparation support was associated with greater attendance intensity.

The posterior evidence profiles caution against a simple account in which media matters only for participation occurrence and support-related practices matter only for participation intensity. Instead, mediated touchpoints may remain relevant among attendees, while support-related practices may mark a social context associated with more intensive participation.

The study contributes to services marketing by separating two aspects of observed participation that are often combined, extending the analysis of media touchpoints beyond participation occurrence and directing attention to practices surrounding the focal customer. Because the evidence is cross-sectional, these conclusions concern conditional association patterns rather than observed customer progression or causal effects.

This component-based framing provides a foundation for longitudinal and experimental research on how mediated contacts, physical encounters and surrounding social practices jointly shape participation in experiential services.

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Table I. Posterior summaries for any live attendance

Variable	Description	Median β	95% CrI	$P(\beta > 0)$	OR	95% CrI for OR
x463	Online viewing: J.League	1.235	0.590 to 1.882	0.999	3.44	1.80 to 6.57
x369	Television viewing: NPB	0.820	0.588 to 1.047	>0.999	2.27	1.80 to 2.85
x573	Transportation support for family members or participants	0.662	0.367 to 0.948	>0.999	1.94	1.44 to 2.58
x460	Online viewing: NPB	0.425	−0.001 to 0.886	0.974	1.53	0.999 to 2.43
x372	Television viewing: J.League	0.314	0.008 to 0.617	0.981	1.37	1.01 to 1.85
x355	Sport-club involvement	0.307	0.010 to 0.606	0.982	1.36	1.01 to 1.83

Source: Author's own work

Note. β denotes the posterior median log-odds coefficient; CrI = credible interval; NPB = Nippon Professional Baseball; OR = odds ratio. The table displays the five coefficients whose 95% CrIs excluded zero in the final main model and online NPB viewing, which is reported transparently because its positive coefficient was similar across specifications although its final main-model 95% CrI narrowly included zero.

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Table II. Posterior summaries for conditional attendance intensity

Variable	Description	Median β	95% CrI	$P(\beta > 0)$	$exp(\beta)$	95% CrI for $exp(\beta)$
x574	Preparation support: drinks or meals for sport participants	0.785	0.380 to 1.171	>0.999	2.19	1.46 to 3.22
x463	Online viewing: J.League	0.703	0.240 to 1.122	0.997	2.02	1.27 to 3.07

Source: Author's own work

Note. β denotes the posterior median log coefficient. $exp(\beta)$ is a ratio on the underlying negative-binomial μ scale before zero truncation and should not be interpreted as a simple observed-count ratio. Conditional response-scale predictions are reported in Table III.

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Table III. Sample-averaged response-scale posterior predictions

Model component	Indicator	Set to 0	Set to 1	Difference	Ratio
Any live attendance	Online J.League viewing	25.7%	48.6%	22.9 percentage points [10.1, 36.0]	—
Conditional attendance intensity	Online J.League viewing	3.95	6.66	2.71 attendances [0.75, 5.37]	1.69 [1.19, 2.37]
Conditional attendance intensity	Preparation support	3.59	6.38	2.79 attendances [1.21, 4.75]	1.77 [1.31, 2.38]

Source: Author's own work

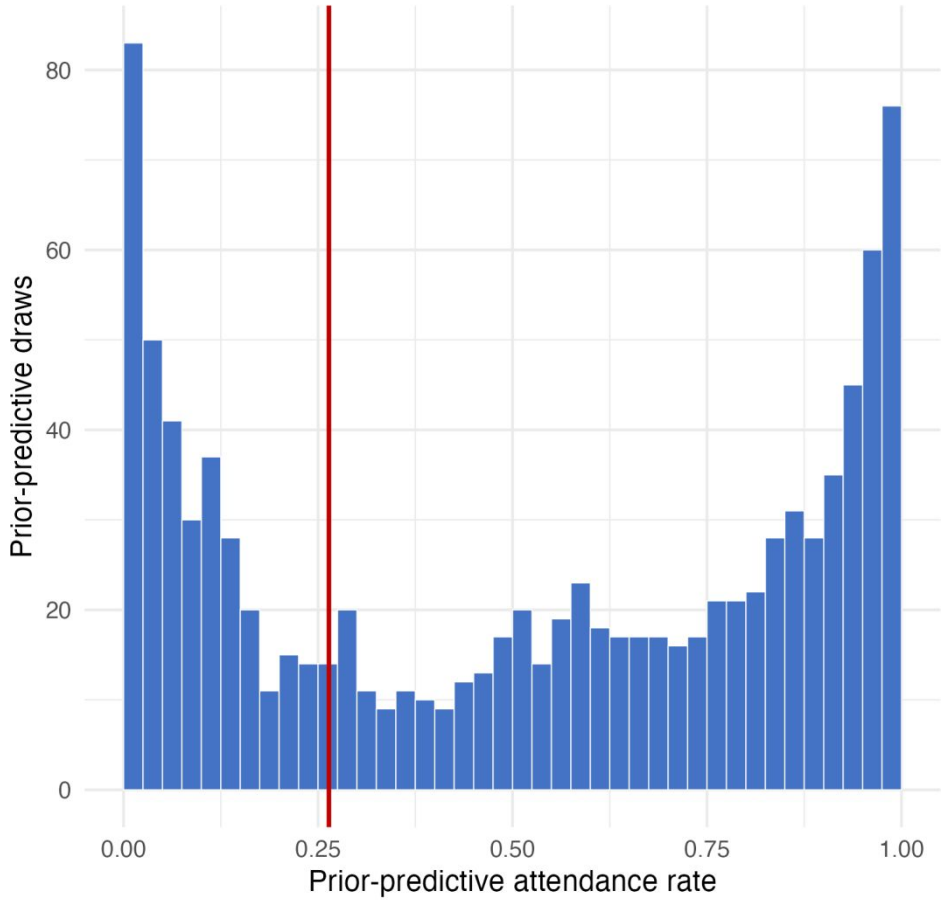
Note. Entries in the Set to 0 and Set to 1 columns are posterior medians. Bracketed values accompanying differences and ratios are 95% credible intervals. For each posterior draw, the indicated binary predictor was set to 0 and 1 for every respondent in the relevant analytic sample, and predictions were averaged over respondents. The conditional-count calculations account for zero truncation. The estimates describe adjusted associations rather than causal effects.

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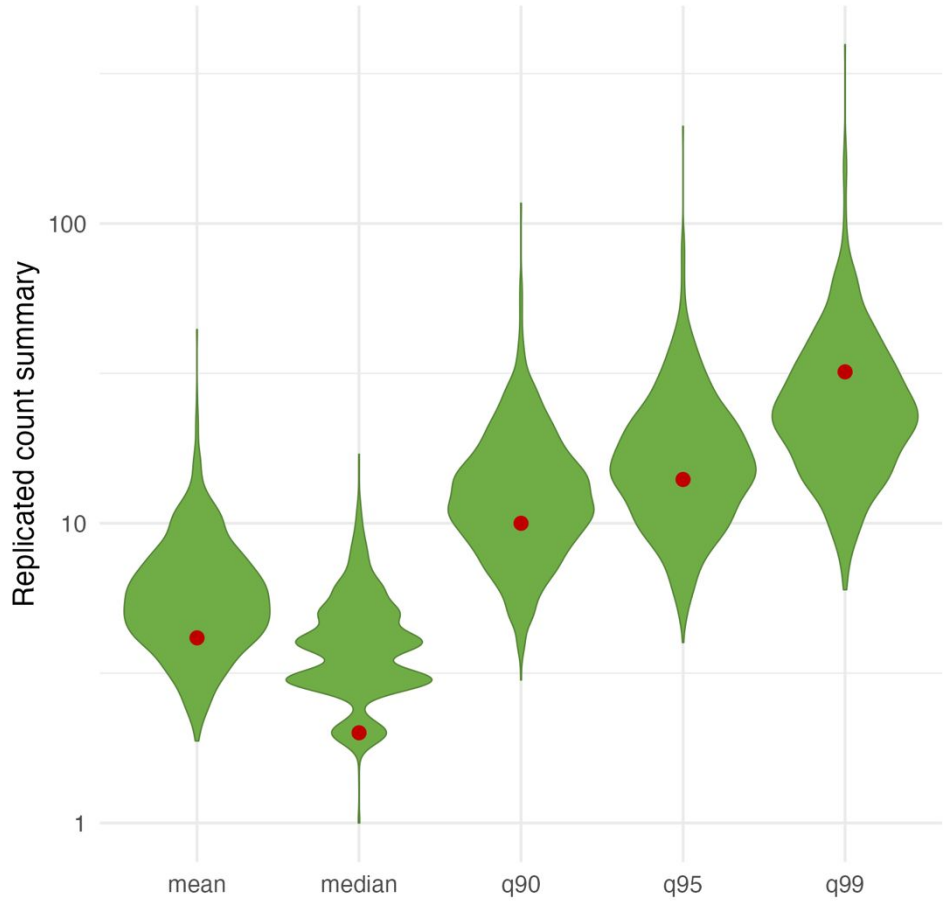
This web appendix reports the prior predictive checks and targeted sensitivity analyses referenced in the Method (Sparse Bayesian estimation) and Results (Response-scale interpretation and sensitivity analyses) of the article.

Figure S1. Prior predictive distributions under the final prior specifications

Panel A. Any live attendance



Panel B. Conditional attendance intensity



Source: Author's own work

Note. Panel A shows the distribution of replicated any-attendance rates generated from 1,000 prior draws; the red vertical line indicates the observed attendance rate. Panel B shows the

distributions of replicated means, medians, and 90th, 95th, and 99th percentiles of conditional attendance intensity; red points indicate the corresponding observed statistics. Each prior draw generated a complete replicated dataset with the same sample size as the corresponding analytic sample (N = 2,887 in Panel A and N = 762 in Panel B). The vertical axis in Panel B is logarithmic. q90, q95, and q99 denote the 90th, 95th, and 99th percentiles, respectively. The count-component prior is data-scaled as described in the Method (Sparse Bayesian estimation) of the article. Prior predictive checks evaluate the implications of the priors and are not posterior model-fit diagnostics.

Table S1. Sensitivity of the reported coefficients to sparsity assumptions and predictor preprocessing

Panel A. Sensitivity to the sparsity assumption

Model component	**Indicator**	**$p_0 = 5$**	**$p_0 = 10$**	**$p_0 = 20$**
Any attendance	Sport-club involvement (x355)	0.307 [0.009, 0.604]; .983	0.307 [0.010, 0.606]; .982	0.307 [0.012, 0.610]; .984
Any attendance	Television viewing: NPB (x369)	0.818 [0.593, 1.039]; >.999	0.820 [0.588, 1.047]; >.999	0.815 [0.595, 1.045]; >.999
Any attendance	Television viewing: J.League (x372)	0.316 [0.014, 0.612]; .985	0.314 [0.008, 0.617]; .981	0.316 [0.010, 0.612]; .983
Any attendance	Online viewing: NPB (x460)	0.432 [0.00003, 0.883]; .975	0.425 [−0.00099, 0.886]; .974	0.428 [0.001, 0.872]; .976
Any attendance	Online viewing: J.League (x463)	1.243 [0.589, 1.880]; >.999	1.235 [0.590, 1.882]; .999	1.233 [0.594, 1.864]; >.999
Any attendance	Transportation support (x573)	0.660 [0.368, 0.939]; >.999	0.662 [0.367, 0.948]; >.999	0.657 [0.364, 0.939]; >.999
Conditional intensity	Online viewing: J.League (x463)	0.708 [0.228, 1.141]; .998	0.703 [0.240, 1.122]; .997	0.697 [0.250, 1.138]; .998
Conditional intensity	Preparation support (x574)	0.785 [0.371, 1.162]; .999	0.785 [0.380, 1.171]; >.999	0.787 [0.379, 1.180]; >.999

Source: Author's own work

Panel B. Sensitivity to predictor preprocessing

Model component	**Indicator**	**Primary preprocessing, $p_0 = 10$**	**Alternative preprocessing, $p_0 = 10$**
Any attendance	Sport-club involvement (x355)	0.307 [0.010, 0.606]; .982	0.323 [0.012, 0.614]; .985
Any attendance	Television viewing: NPB (x369)	0.820 [0.588, 1.047]; >.999	0.816 [0.586, 1.042]; >.999

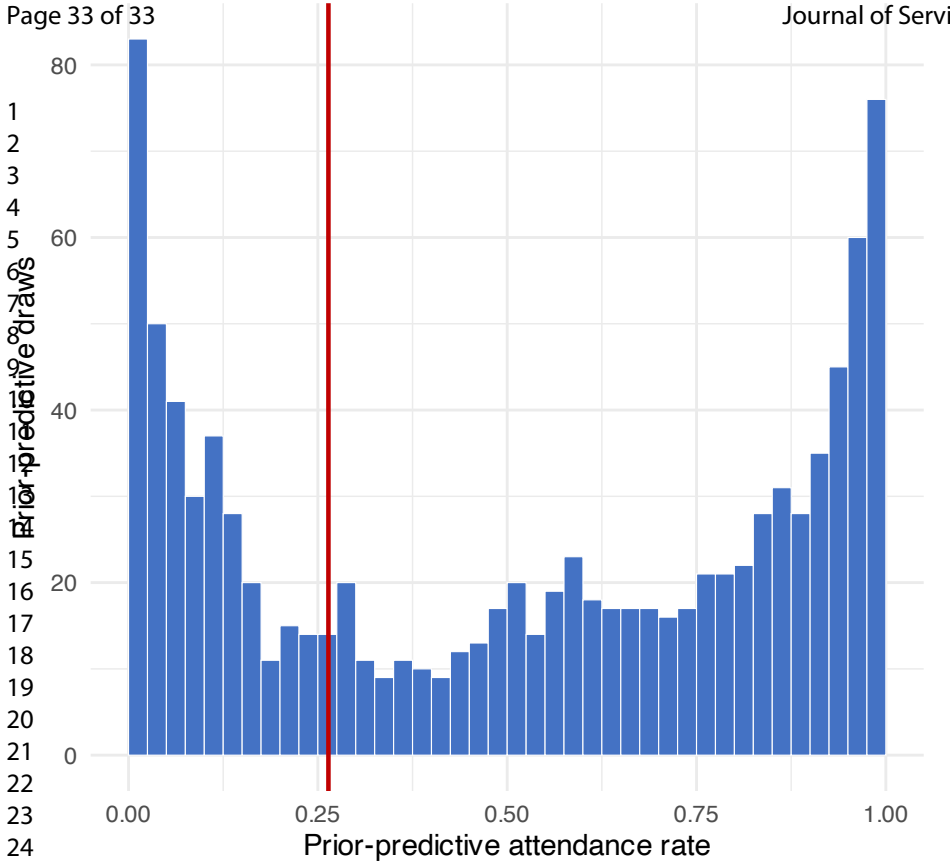
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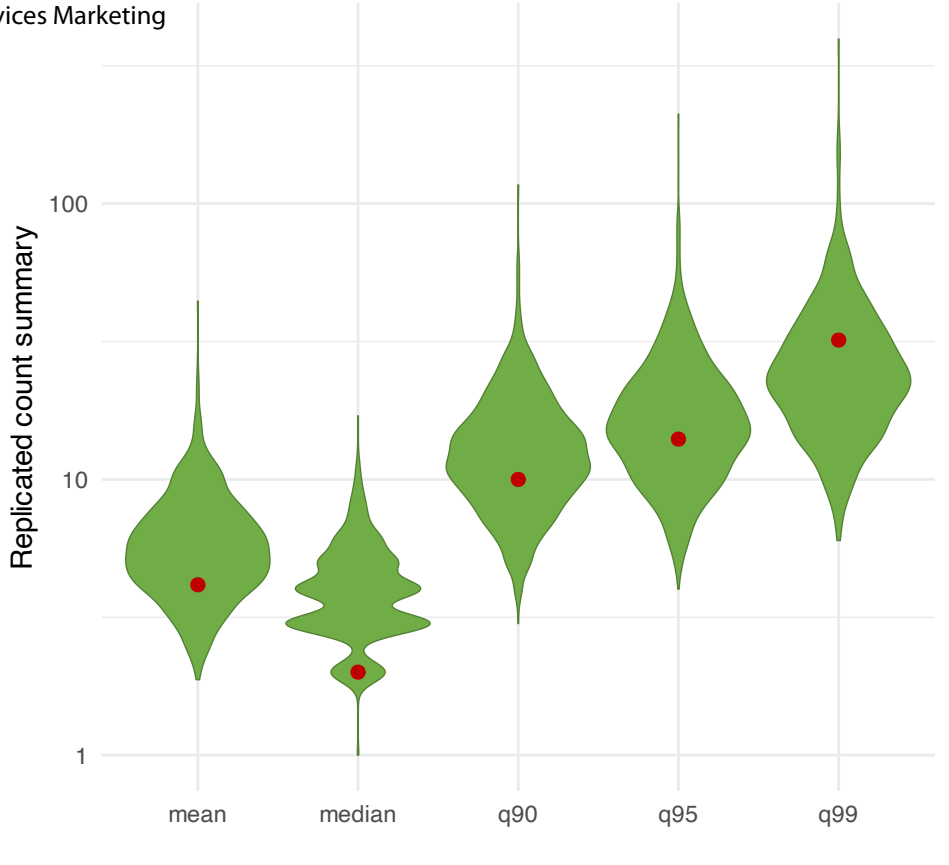
Any attendance	Television viewing: J.League (x372)	0.314 [0.008, 0.617]; .981	0.314 [0.017, 0.610]; .986
Any attendance	Online viewing: NPB (x460)	0.425 [−0.00099, 0.886]; .974	0.434 [0.004, 0.874]; .978
Any attendance	Online viewing: J.League (x463)	1.235 [0.590, 1.882]; .999	1.250 [0.608, 1.883]; >.999
Any attendance	Transportation support (x573)	0.662 [0.367, 0.948]; >.999	0.662 [0.365, 0.941]; >.999
Conditional intensity	Online viewing: J.League (x463)	0.703 [0.240, 1.122]; .997	0.706 [0.230, 1.152]; >.999
Conditional intensity	Preparation support (x574)	0.785 [0.380, 1.171]; >.999	0.801 [0.393, 1.187]; >.999

Source: Author's own work

Note. Entries are posterior medians [95% CrI], followed by the posterior probability of a positive coefficient, P(β > 0 | data). For the any-attendance component, β is a log-odds coefficient. For the conditional-intensity component, β is a log coefficient on the underlying negative-binomial μ scale before zero truncation. CrI = credible interval; NPB = Nippon Professional Baseball. p_0 denotes the expected number of non-negligible coefficients used to calibrate the regularized horseshoe prior. The primary preprocessing standardized continuous and numerically coded ordered predictors by one standard deviation while retaining binary indicators as 0/1. The alternative preprocessing divided continuous predictors by two standard deviations and represented ordered categorical predictors through indicator coding, with the lowest observed category as the reference. The lower credible-interval bounds for online NPB viewing are shown to five decimal places where necessary to avoid obscuring their position relative to zero. Under the alternative preprocessing specification, two additional any-attendance coefficients had 95% CrIs excluding zero: televised rugby viewing (x380), 0.336 [0.009, 0.676], and sport-participation level 2 versus level 1 (x604_L2), 0.323 [0.048, 0.598]. These coding-specific coefficients were not interpreted as principal findings.

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Prior-predictive draws
80
60
40
20
0
0.00
0.25
0.50
0.75
1.00
Prior-predictive attendance rate


Replicated count summary
100
10
1
mean
median
q90
q95
q99